Rethinking case attraction on Ancient Greek infinitive clauses

Caio B. A. Geraldes*
Universidade de São Paulo
<caio.geraldes@usp.br>

June 25, 2025

(1) a. συμβουλεύει **τῷ Ξενοφῶντι ἐλθόντα** εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advices Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)

b. ἀφῆκε **μοι ἐλθόντι** πρὸς ὑμᾶς λέγειν τἀληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

Mentioned works: Lancelot (1658, 1709), Boas et al. (2019), Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013)

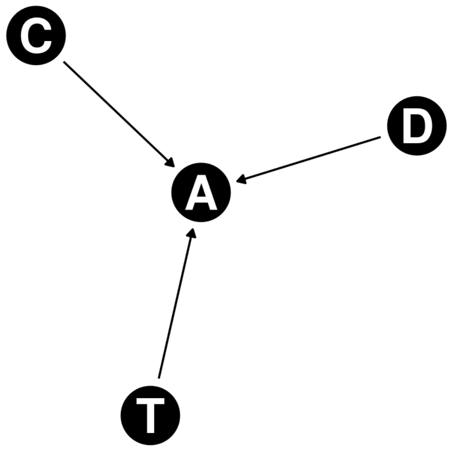
Data source: Diorisis Corpus (Vatri and McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022).

Corpus: Sentences collected from the attic literary sources and Herodotus.

Tooling: R (R Core Team 2013), the packages tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdag (Barrett 2025), and rethinking (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

Typological framework: Corbett (2006)

* This research was funded by the The São Paulo Research Foundation (Fapesp), grant number 2021/06027-4.



Copular infinitives: proposed in by Kühner and Gerth (1898)

(2) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):

$$\begin{array}{l}
\qquad\qquad\qquad\qquad\qquad\qquad\qquad \nearrow \text{Noun} \\
\qquad\qquad\qquad\qquad\qquad \text{Adjective} \\
\qquad\qquad\qquad\qquad \nearrow \\
\qquad\qquad\qquad \text{Participle} \\
\qquad\qquad\qquad \nearrow \\
\text{Attributive} \rightarrow \text{Predicative Verb} \rightarrow \text{Relative Pronoun} \rightarrow \text{Personal Pronoun}
\end{array}$$

$$- \longleftrightarrow +$$

Likelihood of agreement with semantic justification

Impersonal verbs: see Humbert (1945)

Verbs with modal semantics: verbs such as ἐξαρκεῖ and ἔξεστί have higher probability to display case attraction (Geraldes 2020, 2023).

(3) ἀλλ’ ἐξαρκῇ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

Authorship and dialect: Geraldes (2020, 2023) shows that authors use verbs of modal semantic more often in philosophical dialogues.

Word order and Distance: longer distances diminish the likelihood of case attraction. Targets preceding the controller seem to seem to demand case attraction in examples such as (4).

(4) ὡς **ἀγαθοῖς** τε **ὑμῖν** προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

Word order and communicative function: Word Order is not 1-1 equivalent to focus, topic and any other communicative function. It makes an element more likely to be more topical or a focus, but there are unobserved features.

Distance flows from Word Order: Since distance depends on word order, we can only estimate the direct effect by controlling for Word Order.

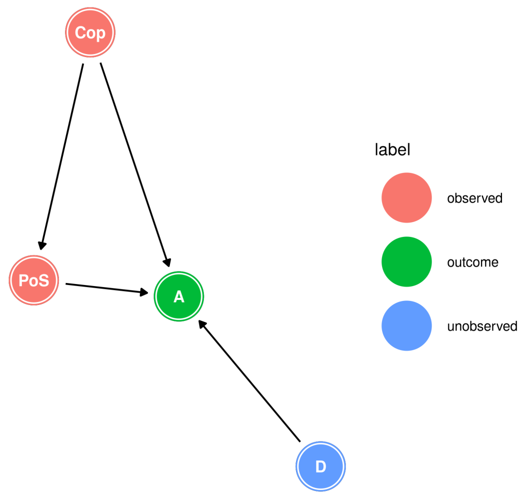


Figure 1: DAG with **A**ttraction, **D**omain, **P**art **o**f **S**preech and **Cop**ula

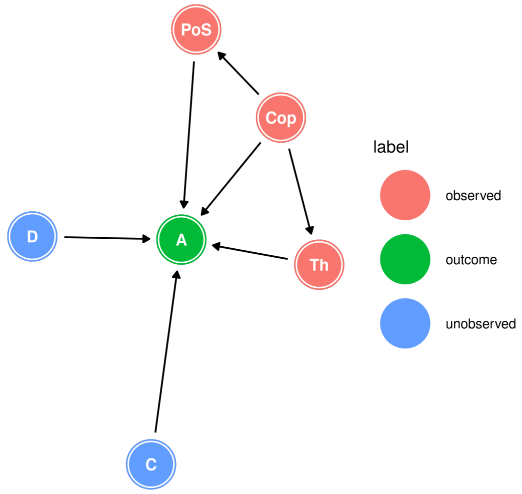


Figure 2: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **of S**preech, **Cop**ula and **Th**ematic role.

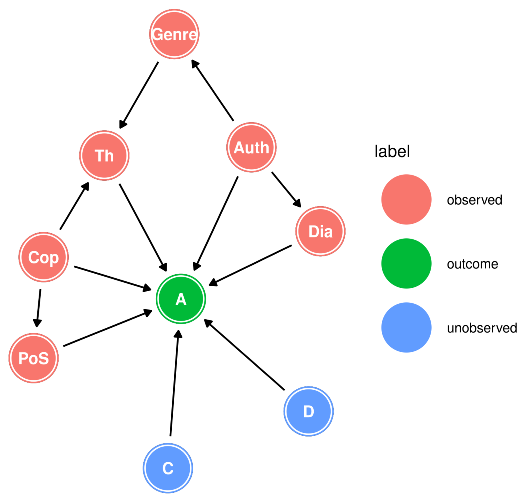


Figure 3: DAG with **A**ttraction, **D**omain, **C**ontroller **P**art **o**f **S**preech, **Cop**ula, **Th**ematic role, **Auth**orship, **Genre**, and **Dia**lect.

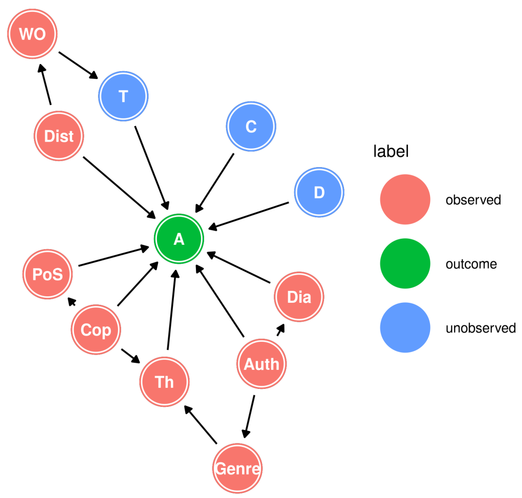
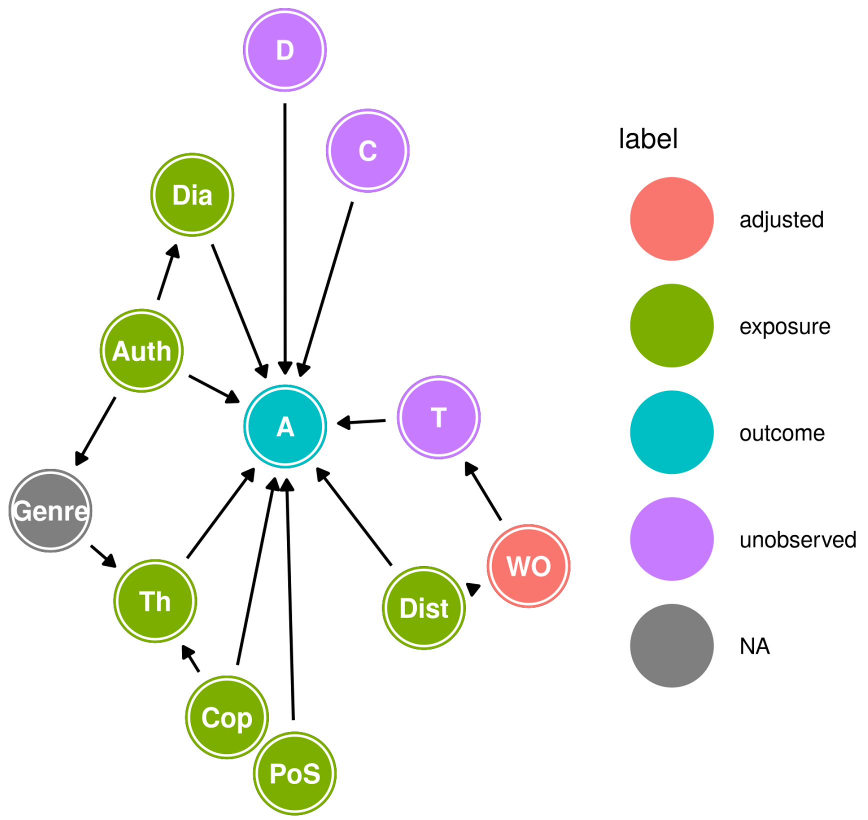


Figure 4: DAG with **A**ttraction, **D**omain, **C**ontroller, **T**arget, **P**art **o**f **S**preech, **Cop**ula, **Th**ematic role, **Auth**orship, **Genre**, **Dia**lect, **Dist**ance, and **W**ord **O**rder.

Final causal model:



Adjustment set: WO

Generalized linear model:

$$\begin{aligned}
\text{A} &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha + \beta_{\text{Cop}}\text{Cop}_i + \beta_{\text{Dist}}\text{Dist}_i + \beta_{\text{WO}}\text{Prec}_i \\
&\quad + \beta_{\text{PoS}} \sum_{j=0}^{PoS_i - 1} \delta_{j_{\text{PoS}}} + \beta_{\text{Th}} \sum_{j=0}^{Th_i - 1} \delta_{j_{\text{Th}}} \\
&\quad + \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} &\sim \text{Normal}(0, 1) \\
\beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{LogNormal}(0, 1) \\
\delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
\beta_{\text{Auth}_j} &= z_{\text{Auth}_j} * \sigma_{\text{Auth}} \\
\beta_{\text{Dia}_j} &= \overline{\beta}_{\text{Dia}} + z_{\text{Dia}_j} * \sigma_{\text{Dia}} \\
\overline{\beta}_{\text{Dia}}, z_{\text{Auth}}, z_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
\sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponential}(1)
\end{aligned}$$

Copula and Distance

Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	2.03	0.39	1.42	2.66	1
β_{Dist}	-0.22	0.04	-0.29	-0.16	1

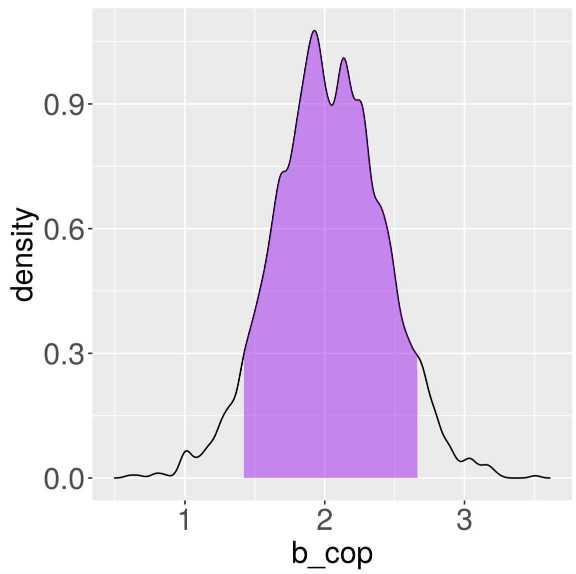


Figure 5: Posterior distribution of β_{Cop} with the 89% Highest Posterior Density Interval.

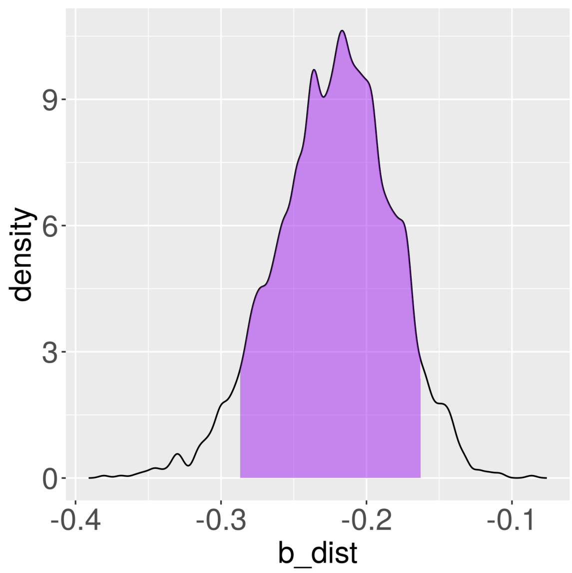


Figure 6: Posterior distribution of β_{Dist} with the 89% Highest Posterior Density Interval.

Part of Speech and the Predicate Hierarchy

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	0.45	0.28	0.11	0.95	1
δ_{Adj}	0.53	0.18	0.23	0.82	1
δ_{Noun}	0.47	0.18	0.18	0.77	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	$\beta_{\text{PoS}} * 0$	0.00	0.00	0.00	0.00	NA
Adjective	$\beta_{\text{PoS}} * \delta_{\text{Adj}}$	0.25	0.18	0.05	0.60	1
Noun	$\beta_{\text{PoS}} * (\delta_{\text{Adj}} + \delta_{\text{Noun}})$	0.45	0.28	0.11	0.95	1

Thematic Role of the Controller

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	2.47	0.40	1.84	3.12	1
δ_{Exp}	0.34	0.11	0.17	0.51	1
δ_{Agent}	0.66	0.11	0.49	0.83	1

Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	$\beta_{\text{Th}} * 0$	0.00	0.0	0.00	0.00	NA
Experiencer	$\beta_{\text{Th}} * \delta_{\text{Exp}}$	0.83	0.3	0.38	1.34	1
Agent	$\beta_{\text{Th}} * (\delta_{\text{Exp}} + \delta_{\text{Agent}})$	2.47	0.4	1.84	3.12	1

Dialect and Authorship

Effect	mean	sd	5.5%	94.5%	rhat
$\overline{\beta}_{\text{Dia}}$	-0.49	0.84	-1.78	0.90	1.00
σ_{Dia}	1.20	0.78	0.30	2.67	1.01
β_{Attic}	-0.16	0.89	-1.55	1.28	1.00
β_{Jonic}	-1.59	1.01	-3.19	-0.02	1.00
$\beta_{\text{Attic}} - \beta_{\text{Jonic}}$	1.43	0.69	0.26	2.49	NA

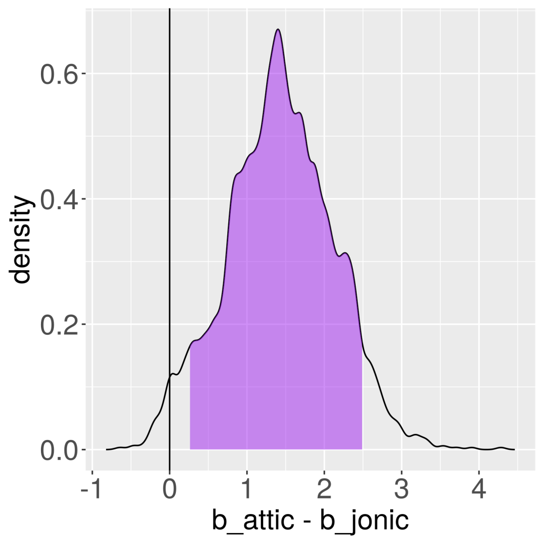
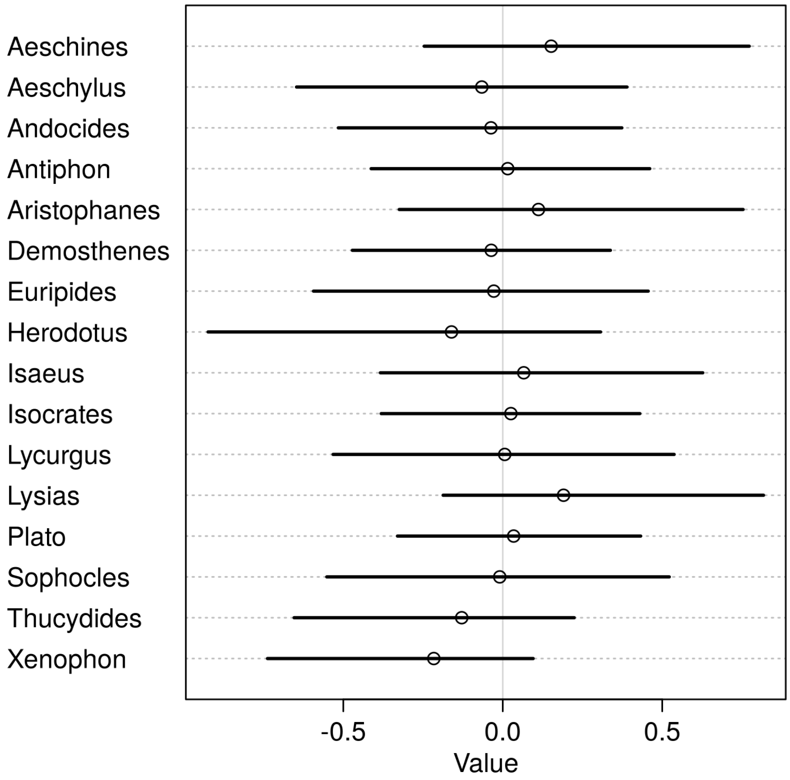
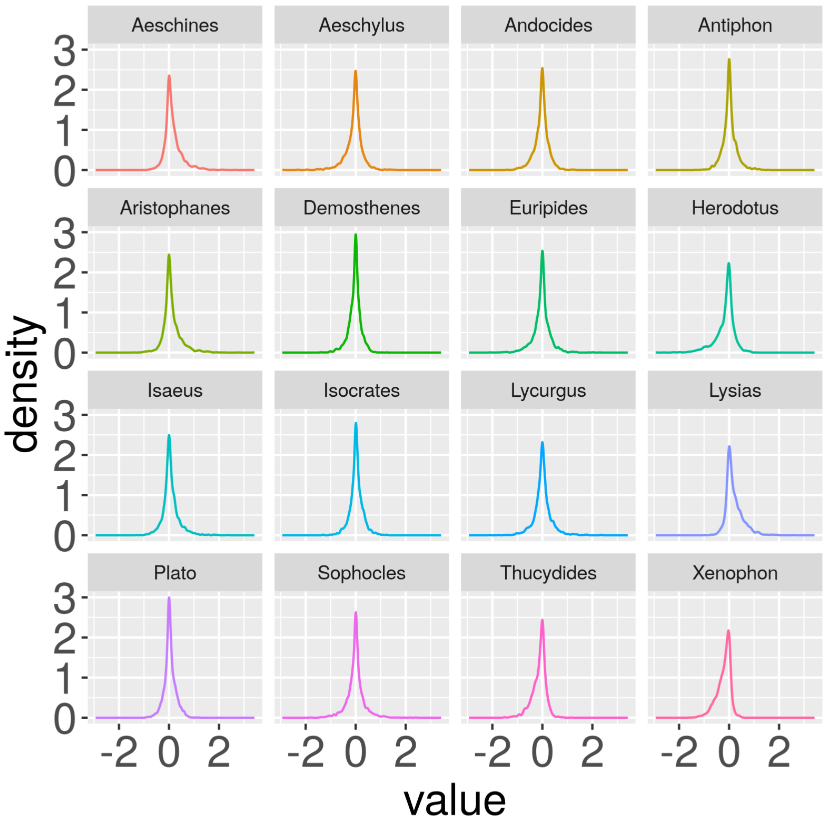


Figure 7: Posterior distribution of the difference between β_{Attic} and β_{Jonic} with the 89% Highest Posterior Density Interval.


Aeschines
Aeschylus
Andocides
Antiphon
Aristophanes
Demosthenes
Euripides
Herodotus
Isaeus
Isocrates
Lycurgus
Lysias
Plato
Sophocles
Thucydides
Xenophon
-0.5
0.0
0.5
Value


Aeschines
Aeschylus
Andocides
Antiphon
Aristophanes
Demosthenes
Euripides
Herodotus
Isaeus
Isocrates
Lycurgus
Lysias
Plato
Sophocles
Thucydides
Xenophon
density
0
1
2
3
-2
0
2
value

References

ANDREWS, A. D. (1971). "Case Agreement of Predicate Modifiers in Ancient Greek". In: *Linguistic Inquiry* 2.2, pp. 127–151.

BARRETT, M. (2025). *ggdag: Analyze and Create Elegant Directed Acyclic Graphs*. R package version 0.2.13.9000. URL: https://github.com/r-causal/ggdag.

BOAS, E. v. E. et al. (2019). *The Cambridge Grammar of Classical Greek*. Cambridge: Cambridge University Press.

CORBETT, G. G. (2006). *Agreement*. Cambridge: Cambridge University Press.

GERALDES, C. B. A. (2020). " Case Attraction on Infinitive Clauses of Ancient Greek: A case study on Herodotus, Plato and Xenophon ". MA thesis. São Paulo: Universidade de São Paulo. DOI: https://doi.org/10.11606/D.8.2020.tde-12042021-174449.

— (2023). "Atração infinitiva, assimilação, transmissão ou concordância: Notas sobre a literatura em Grego Antigo". In: *Revista Letras* 108. DOI: 10.5380/rel.v108i0.90501.

HUMBERT, J. (1945). *Syntaxe grecque*. Paris: C. Klincksieck.

KÜHNER, R. and B. GERTH (1898). *Ausführliche Grammatik der griechischen Sprache*. Hannover: Hahn.

LAKOFF, G. (1970). "Global rules". In: *Language* 46.3, pp. 627–639.

LANCELOT, C. (1658). *Nouvelle Methode pour apprendre facilement la Langue Greque: contenant les regles de Déclinaisons, des Conjugaisons, de l' Investigation du Theme, de la Syntaxe, de la Quantité, des Accens, des Dialectes, & des Licences Poëtiques*. Paris: Pierre le Petit.

— (1709). *Nouvelle Methode pour apprendre facilement la Langue Latine: contenant les regles de Genres, des Déclinaisons, des Préterits, de la Syntaxe, de la Quantité, & des Accens*. Paris: Denys Marietti.

MCELREATH, R. (2020). *Statistical Rethinking*. Boca Raton: Taylor & Francis Group, LLC.

QUICOLI, A. C. (1982). *The Structure of Complementation*. Ghent: E. Story-Scientia.

R CORE TEAM (2013). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria. URL: http://www.R-project.org/.

SEVDALI, C. (2007). " Infinitival clauses in Ancient Greek: overt and null subjects, the role of case and focus ". PhD thesis. Cambridge: University of Cambridge.

— (2013). "Case Transmission beyond Control and the Role of Person". In: *Journal of Historical Syntax* 2.4, pp. 1–52.

SPYROPOULOS, V. (2005). "The syntax of Classical Greek infinitive". In: *Universal Grammar in the Reconstruction of Ancient Languages*. Ed. by K. KISS. Berlin: Mouton de Gruyter, pp. 295–337.

STAN DEVELOPMENT TEAM (2025a). *Stan Reference Manual*. Version v2.37.0-rc2. URL: https://mc-stan.org.

— (2025b). *Stan User's Guide*. Version v2.37.0-rc2. URL: https://mc-stan.org.

TANTALOU, N. (2003). "Infinitives with overt subjects in Classical Greek". In: *Μελέτες για την ελληνική γλώσσα: πρακτικά της 23ης ετήσιας συνάντησης του τμήματος φιλολογίας του Αριστοτελείου Πανεπιστημίου Θεσσαλονίκης, 17-19 Μαΐου, 2002*. Θεσσαλονίκη: Αφοί Κυριακίδη, pp. 358–365.

TEXTOR, J. et al. (2016). "Robust causal inference using directed acyclic graphs: the R package 'dagitty'". In: *International Journal of Epidemiology* 6.45, pp. 1887–1894.

VATRI, A. (2020-2022). *Diorisis Search 3.11*. Version 3.11. URL: https://www.crs.rm.it/diorisissearch/ (visited on 02/28/2023).

Vatri, A. and B. McGillivray (2018). “The Diorisis Ancient Greek Corpus”. In: *Research Data Journal for the Humanities and Social Sciences* 3.1, pp. 55–65. URL: https://brill.com/view/journals/rdj/3/1/article-p55_55.xml.

— (2020). “ Lemmatization for Ancient Greek: An experimental assessment of the state of the art ”. In: *Journal of Greek Linguistics* 20.2, pp. 179–196. DOI: https://doi.org/10.1163/15699846-02002001.

Wickham, H. et al. (2019). “Welcome to the tidyverse”. In: *Journal of Open Source Software* 4.43, p. 1686. DOI: 10.21105/joss.01686.